

The **Dirac delta function** (often written as $\delta(x)$) is a mathematical object used to model an **idealized point effect**-something that is zero everywhere except at one point, yet has a finite total “strength.”

It’s not a function in the usual sense; it’s a **distribution (generalized function)**.

Intuitive idea

Think of $\delta(x)$ as an **infinitely tall, infinitely narrow spike at $x = 0$** whose total area is 1.

Defining properties

1. Zero everywhere except at 0

$$\delta(x) = 0 \quad \text{for } x \neq 0$$

2. Unit area

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

3. Sifting (sampling) property

For any “nice” function $f(x)$,

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$$

This is the most important property: **the delta function “picks out” the value of a function at a point.**

Shifted delta function

$$\delta(x - a)$$

represents a spike at $x = a$ instead of 0.

Scaling property

$$\delta(ax) = \frac{1}{|a|} \delta(x)$$

This ensures the area remains 1 even when the spike is compressed or stretched.

Physical meaning

The Dirac delta is used to represent **point sources**:

- **Physics**
 - Point mass: $\rho(x) = m \delta(x - x_0)$
 - Point charge in electromagnetism
 - Impulse force in mechanics
 - **Signal processing**
 - Ideal impulse signal
 - System impulse response
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Relation to ordinary functions

You can think of $\delta(x)$ as the limit of increasingly narrow functions with unit area, such as:

$$\delta(x) = \lim_{\sigma \rightarrow 0} \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}$$

(a Gaussian approximation)

Derivative connection

The delta function is the derivative of the **Heaviside step function** $H(x)$:

$$\frac{d}{dx} H(x) = \delta(x)$$

Key takeaway

- The Dirac delta is **not a real function**
- It represents an **idealized point effect**
- Its power lies in integration, not pointwise values

If you want, I can explain it **visually**, **in physics terms**, or **rigorously using distributions**.